

Abdullah (Arby) Abood

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PROFESSIONAL SUMMARY

Senior data scientist/computational biologist with experience building regulated-life-sciences analytics, AI-enabled evidence systems, predictive models, and reproducible pipelines. Strong record translating complex clinical, multi-omics, and product-quality data into traceable decision-support tools for Regulatory, Clinical, Product Development, and R&D stakeholders.

WORK EXPERIENCE

COLGATE-PALMOLIVE | SENIOR SCIENTIST

Research Triangle Park, NC | Jun 2023 – Present

- Architected a Neo4j-backed GraphRAG pipeline that ingests unstructured final study reports on HPC, applies open-source parsing, embedding, and NER models, and retrieves evidence from third-party knowledge graphs to support traceable, lower-hallucination scientific question answering.
- Co-led development of an AI agent that generated draft nutritional-claims substantiation packages estimated at ~80% completion by grounding outputs in AAFCO regulations, Hill's Global Nutritional Standards, and 12K historical Veeva claims. Reduced review cycle time by ~50% in pilot workflows and standardized audit-ready claim templates across NA, EU, and global regulatory teams.
- Built disease prediction models using XGBoost and Random Forest, along with biological age estimation models using LASSO-penalized Cox proportional hazards and Gompertz approaches, on CBC/chemistry panel data from ~1,000 clinical cases. Achieved PR-AUC up to 0.92 and enabled early risk detection and safety-relevant stratification across clinical populations.
- Developed reproducible Nextflow pipelines for bulk RNA-seq, single-cell RNA-seq, and whole-genome bisulfite sequencing across short- and long-read data, improving QC standardization, portability, and run-to-run reproducibility on HPC.
- Partnered with analytical scientists to design and analyze metabolomics stability experiments using repeated-measures ANOVA with orthogonal polynomial contrasts for linear, quadratic, and cubic trends, delivering decision-ready summaries and dashboards for leadership review and product-quality decisions.

UNIVERSITY OF VIRGINIA | COMPUTATIONAL BIOLOGIST

Charlottesville, VA | Jul 2018 – May 2023

- Identified causal isoforms in bone mineral density GWAS by integrating sQTL data and Bayesian analysis, refined with long-read RNA-seq and proteomics, leading to the implication of genes in osteoporosis through *in vitro* validation.
- Uncovered long non-coding RNAs impacting BMD GWAS loci through eQTL analysis, Bayesian colocalization, TWAS, and allelic imbalance, revealing key non-coding elements of the bone transcriptome and their roles in osteoporosis.
- Built reproducible computational workflows for multi-omics integration, variant-to-function interpretation, and disease-mechanism prioritization across transcriptomic, proteomic, and genetic datasets.

SELECTED PUBLICATIONS

- **Abood A** et al. Long-read proteogenomics to connect disease-associated sQTLs to the protein isoform effectors of disease. *American Journal of Human Genetics*, 2024.
- Castaldi P, **Abood A** et al. Bridging the splicing gap in human genetics with long-read RNA sequencing. *Human Molecular Genetics*, 2022.
- **Abood A** et al. Identification of known and novel long non-coding RNAs potentially responsible for the effects of BMD GWAS loci. *Journal of Bone and Mineral Research*, 2022.
- **Abood A**, Farber CR. Using “-omics” Data to Inform Genome-wide Association Studies in the Osteoporosis Field. *Current Osteoporosis Reports*, 2021.

TECHNICAL SKILLS

Languages & Tools: Python, R, SQL, Git, Neo4j, Docker, HPC/SLURM, Nextflow

ML & Statistics: XGBoost, Random Forest, Logistic Regression, Cox Proportional Hazards, ANOVA, DOE, Mixed Models, Survival Analysis

AI & NLP: LLM/RAG/GraphRAG systems, NER/NLP, embeddings, unstructured data processing, HuggingFace

Life Sciences Analytics: Multi-omics integration, clinical data analysis, evidence synthesis, data quality, reproducible pipelines, regulatory and claims support

SPECIALIZED TRAINING

BIOMEDICAL DATA SCIENCE TRAINING GRANT

Awarded NIH T32 Biomedical Data Science Training Grant supporting advanced research in multi-omics integration and high-dimensional clinical data analysis to advance understanding of disease mechanisms.

EDUCATION

Ph.D. COMPUTATIONAL BIOLOGY

Charlottesville, VA | May 2023

UNIVERSITY OF VIRGINIA

Teaching Assistant (TA): Core Course in Integrative Biosciences (BIMS 6000), Recent Advances in Public Health Genomics (PHS 5705)